

Dhyani Soni

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EDUCATION

NEW JERSEY INSTITUTE OF TECHNOLOGY

B.S. in Computer Science & AI Minor, Albert Dorman Honors College

Expected May 2027

Newark, NJ

GPA: 3.62 (Dean's List) | Relevant Coursework: Data Structures & Algorithms, Artificial Intelligence, Data Science, Database Systems, Computer Networks, Operating Systems, Cybersecurity, Linear Algebra, AI-Assisted Software Engineering, Data Visualization

- **Campus Involvement:** Lyra STEM Curriculum Manager, Kids Who Code Treasurer, SHPE, Rewriting the Code (RTC)

SKILLS

- **Languages:** Python, SQL, Java, JavaScript, C/C++, HTML/CSS
- **Frameworks & Libraries:** React, FastAPI, Flask, Spring Boot, Streamlit, TensorFlow, PyTorch, Scikit-learn, YOLO, FAISS, NumPy, Pandas, Matplotlib, Tailwind CSS
- **AI/ML:** RAG, Prompt Engineering, Multi-Agent Systems, LangChain, OpenCV, Amazon Bedrock
- **Cloud & DevOps:** AWS (S3, Lambda, DynamoDB, API Gateway), Docker, Terraform, GitHub Actions, CI/CD
- **Tools:** Git, Jupyter Notebook, Tableau, Figma, Postman, MySQL, Anaconda, GitHub Copilot, MS Office

PROFESSIONAL EXPERIENCE

Agentic AI Engineering Intern @ Tata Consultancy Services - TCS

May 2026 - Aug 2026

- Designing a **multi-step AI agent** on AWS Bedrock that **validates stored procedure** outputs against a 20-point requirements report checklist, **flagging** data inaccuracies and **automating** a previously manual QA process
- Architecting serverless infrastructure using Lambda, API Gateway, S3, DynamoDB, and Terraform to support scalable cloud deployments

Artificial Intelligence and Machine Learning Fellow

Mar 2026 - Mar 2027

Break Through Tech @ Cornell Tech

- Selected nationally from **4,000+** applicants for a rigorous year long **AI Fellowship at Cornell Tech**
- Developing **hands-on machine learning expertise** through an accelerated **Python bootcamp** and advanced coursework in data wrangling, predictive modeling, and performance evaluation, applying **Pandas, Scikit-Learn, and Matplotlib**
- Scheduled to lead an **AIStudio** project with an **industry sponsor** (Aug 2026–Mar 2027), managing a cross-functional team to ship a **deployment-ready ML product** and deliver final outcomes to **senior executive leadership**

Artificial Intelligence and Machine Learning Research Intern

Sept 2025 - May 2026

CNALab @ New Jersey Institute of Technology

- Won **1st Place** at NJIT's Honors Interdisciplinary Research Forum (**40+ competitors**) by delivering an interdisciplinary research presentation that clearly connected methods to **real-world impact**; also selected as a **Finalist** for the Undergraduate Research and Innovation Program.
- Developed an AI-powered rehabilitation tool for **patient mobility tracking** in aquatic therapy by implementing pose estimation with **OpenCV** and **Python**; built a custom preprocessing pipeline with **image enhancement**
- **Improved keypoint detection by 35%** and reduced labeling errors in the training dataset, increasing joint torque and acceleration analysis accuracy underwater

UX Research Assistant

May 2024 - Dec 2024

Department of Informatics @ New Jersey Institute of Technology

- Led qualitative UX research for **inclusive interface design**; conducted and analyzed 25+ interviews with older adults to uncover **accessibility gaps in digital safety**
- Converted behavioral insights into actionable recommendations via 3 designs of UX **storyboards** and **strategy reports**
- Applied **market research techniques** like user personas, segmentation analysis, and journey mapping to inform CX and product strategies in customer-facing roles, influencing inclusive **UI changes** for campus-wide platforms

PROJECTS

Nexus | JavaScript, React, Flask, SQLite, Jest, Pytest, Stryker, Render

Mar-May 2026

- Built and deployed a LinkedIn-style full-stack application with a Flask REST API and **AI-powered outreach message guidance, recruiter mode, conference discovery** and **WCAG-compliant** keyboard navigation support
- Achieved **96% backend route coverage** and **69.41% mutation score** across **1,010 mutants** using Jest, Pytest, and Stryker mutation testing
- Configured **GitHub Actions CI/CD pipeline** and a **protected main branch**, enabling zero-touch backend deploys to Render on every merge

SenseLense | Python, SQLite, ElevenLabs, Presage, React

Feb 2026

- **Won Best AI Analytics Track at HackHers Hackathon (ADP Challenge)** by architecting a real-time multimodal sentiment analysis pipeline for sales pitch evaluation at **ADP**
- Integrated Presage (emotion detection), ElevenLabs (speech-to-text), and a **timestamped database** to structure client interaction data
- Leveraged Gemini for **contextual mood analysis** to flag **real-time sentiment shifts** during live pitches, **notifying** reps of negative engagement and suggesting **optimized high-impact pitch strategies**

AgentWeave | Python, FastAPI, YOLOv5, FAISS, CLIP, React, Firebase

July 2025

- Engineered a scalable AI-driven **clothing recommendation** tool using YOLOv5 and CLIP to detect users and generate outfit matches based on visual similarity
- Deployed **FAISS vector** search for real-time outfit retrieval, and built a **swipe-based user interface in React** to collect behavioral feedback and write preference data directly to Firebase for usage analytics
- Clustered user interactions with embedding vectors to **dynamically refine** model accuracy and **personalize recommendations**